

AI ARCHITECT — PREDICTIVE PORT OPERATIONS CONTROL CENTRE

Integrated Port Congestion Prediction,
Resource Optimisation & 72-Hour Planning

IBM BoB AI Innovation Hackathon 2026

Predict congestion before it becomes a queue.
Optimise resources. Act before delays cascade.

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PORT CONGESTION IS MANAGED TOO LATE

Manual planning and spreadsheets can't keep up with real-time complexity

Congestion becomes visible only after vessels wait and conflicts form

2021 Los Angeles/Long Beach backlog: 100+ ships waiting offshore for weeks

\$10B+ impact on global supply chains

Can supervisors predict congestion and act before it becomes critical?



PREDICTIVE PORT OPERATIONS CONTROL CENTRE

A unified cockpit for port operations and shift supervisors

PREDICT → EXPLAIN → OPTIMIZE → ACT → PLAN → SIMULATE

Vessel intelligence and congestion risk in one view

Optimise berth and crane assignments

Plan 72 hours of operations with clarity

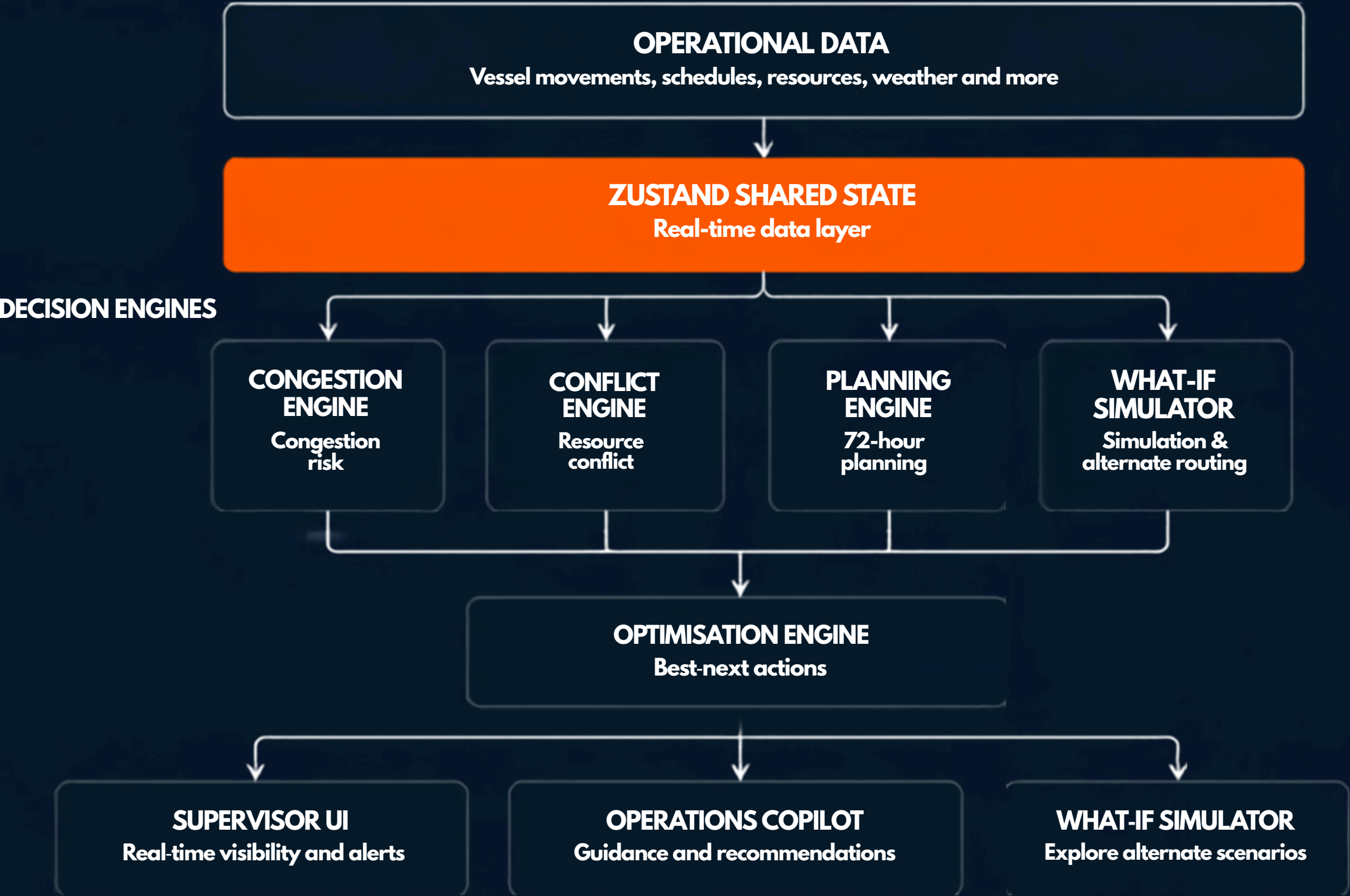
Simulate alternatives before committing

AI copilot summarises, guides, and recommends



HOW IT WORKS

Predict, plan and act before congestion hits



TECHNOLOGY STACK

IBM Bob IDE
Core development environment

React + TypeScript + Vite
Frontend

Tailwind CSS + Lucide React UI

Zustand
State management

Recharts
Analytics

DECISION ENGINES COVER

Congestion Risk
Resource Conflict
Optimisation
72-Hour Planning
What If Simulation
Alternate Routing
Operations Copilot

CURRENT POC NOTE

Uses transparent deterministic decision-support logic with simulated operational data

WHAT-IF → OPTIMISE → APPLY

simulate disruptions, optimise outcomes, act with confidence

TEST SCENARIOS BEFORE THEY IMPACT OPERATIONS

- Vessel Delay: +2 hours
- Crane Outage: extended maintenance
- Early Vessel Arrival: -3 hours



OPTIMISATION METRICS

METRIC	BEFORE	AFTER
Waiting Time	[X]	[Y]
Conflicts	[X]	[Y]
Crane Shortfalls	[X]	[Y]
Berths at Risk	[X]	[Y]
Average Utilisation	[X%]	[Y%]



“ Instead of reacting to disruption, supervisors can simulate it, evaluate the impact and **apply a safer operational plan.** ”

BUILT WITH IBM BOB IDE

Develop, integrate, and harden faster

Transparent logic builds supervisor trust

Future-ready with watsonx.ai for richer explanations

An open foundation for continuous innovation



Built for trust

Transparent decision logic
enables easier understanding
and supervisor trust

Built for the future

watsonx.ai integration
will enable richer natural-language
explanations and decision-support

FROM REACTIVE OPERATIONS TO PROACTIVE DECISION SUPPORT

PROTOTYPE AT A GLANCE



EXPECTED OPERATIONAL VALUE

- Earlier Risk Detection
- Better Resource Utilisation
- Reduced Waiting
- Faster Decision-Making
- Safer Disruption Handling

Data is simulated prototype data, not live port information



AI ARCHITECT

THE TEAM BEHIND PREDICTIVE PORT OPERATIONS

**PRIT
TEJANI**

**FRONTEND • UI/UX
+ INTEGRATION**

Control-centre interface
and dashboards

Module integration

User workflow
optimization

**SAVAN
SOJITRA**

AI/ML ENGINEER

Demand forecasting
and congestion
prediction models

Real-time anomaly
detection and alerts

Model monitoring
and continuous
improvement

**ADITYA
AMIPARA**

DATA ENGINEER

Data pipelines
and integration

Data quality
and governance

Scalable data
infrastructure

**JAHNAVI
SUTHAR**

PRODUCT ANALYST

Business requirements
and use cases

KPI definition and
impact tracking

Stakeholder alignment
and insights

“ We don't just show the supervisor what is happening. We help them **decide what to do next.** ”